

Revisiting topological fixed point theory

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1 Fixed point theory and six-functor formalisms

Throughout this section, let $(\mathcal{C}, \mathcal{E})$ denote a geometric setup equipped with a presentable six-functor formalism

$$\mathcal{D}': \mathrm{Span}(\mathcal{C})_{\mathcal{E}} \longrightarrow \mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$$

in the sense of [Man22, App A.5]. We assume for simplicity that $\mathcal{C}$ contains all finite limits, so that $\mathcal{D}'$ is a lax symmetric monoidal functor, where $\mathrm{Span}(\mathcal{C})_{\mathcal{E}}$ is equipped with the symmetric monoidal structure described in [Man22, Rem A.5.5]. It follows formally that $\mathcal{D}'(\mathrm{pt})$ has a canonical commutative algebra structure and that $\mathcal{D}'$ factors through a symmetric monoidal functor

$$\mathcal{D}: \mathrm{Span}(\mathcal{C})_{\mathcal{E}} \longrightarrow \mathrm{Pr}_{\mathcal{D}'(\mathrm{pt})}^{\mathrm{L}}.$$

(Note that $\mathcal{D}(X)$ has the same underlying presentable stable ∞ -category as $\mathcal{D}'(X)$.) We will assume the following:

- (A) The functor $\mathcal{D}$ is a symmetric monoidal functor, i.e. the canonical functor

$$\mathcal{D}(X) \otimes_{\mathcal{D}(\mathrm{pt})} \mathcal{D}(Y) \longrightarrow \mathcal{D}(X \times Y)$$

is an equivalence for each pair of objects X and Y in $\mathcal{C}$.

Notation 1.1. For the remainder of this section, we let $\mathcal{A} = \mathcal{D}(\text{pt})$.

Assumption (A) is satisfied for the six-functor formalisms that interest us in this paper, such as the six-functor of sheaves of on locally compact Hausdorff spaces and of local systems on anima.

Find reference for the fact that it fails for étale sheaves

1.1 The categorified Lefschetz trace formula

The starting point is the following observation about dualizable objects in span categories:

Lemma 1.2. *If $X \in \mathcal{C}$ is admissible, then $X \in \text{Span}(\mathcal{C})_{\mathcal{E}}$ is self-dual (and in particular dualizable), with evaluation and unit given by the spans*

$$X \xleftarrow{\text{id}} X \xrightarrow{\Delta} X \times X, \quad \text{pt} \leftarrow X \xrightarrow{\text{id}} X$$

respectively, and counit and coevaluation given by the flipped spans.

Using our assumption (A), we deduce that $\mathcal{D}(X)$ is dualizable as an object of $\text{Pr}_{\mathcal{A}}^{\text{L}}$ for each admissible $X \in \mathcal{C}$. In particular, for an endofunctor $F: \mathcal{D}(X) \rightarrow \mathcal{D}(X)$, we can take the trace $\text{tr}_{\text{Pr}_{\mathcal{A}}^{\text{L}}}(F \curvearrowright \mathcal{D}(X))$. The first incarnation of the Lefschetz trace formula that we consider is a description of this trace when F is a pullback functor f^* for some endomorphism $f \curvearrowright X$ in $\mathcal{C}$.

Proposition 1.3 (Categorified trace formula). *Let $\mathcal{D}: \text{Span}(\mathcal{C})_{\mathcal{E}} \rightarrow \text{Pr}_{\mathcal{A}}^{\text{L}}$ be a symmetric monoidal, $\mathcal{A}$ -linear six-functor formalism and let $X \in \mathcal{C}$ be admissible. For every endomorphism $f: X \rightarrow X$, we have that*

$$\text{tr}_{\text{Pr}_{\mathcal{A}}^{\text{L}}}(f^* \curvearrowright \mathcal{D}(X)) \simeq (X^f)_!(X^f)^* \in \text{Map}_{\text{Pr}_{\mathcal{A}}^{\text{L}}}(\mathcal{A}, \mathcal{A}), \quad (1.1)$$

where $X^f \in \mathcal{C}$ is defined by the pullback

$$\begin{array}{ccc} X^f & \xrightarrow{\quad} & X \\ \downarrow & \lrcorner & \downarrow \Delta \\ X & \xrightarrow{(\text{id}, f)} & X \times X. \end{array} \quad (1.2)$$

If additionally $f: X \rightarrow X$ is admissible, we moreover have

$$\text{tr}_{\text{Pr}_{\mathcal{A}}^{\text{L}}}(f! \curvearrowright \mathcal{D}(X)) \simeq (X^f)_!(X^f)^*. \quad (1.3)$$

Proof. Recall that f^* is the result of applying the symmetric monoidal functor $\mathcal{D}: \text{Span}(\mathcal{C})_{\mathcal{E}} \rightarrow \text{Pr}_{\mathcal{A}}^{\text{L}}$ to the span

$$\begin{array}{ccc} & Y & \\ f \swarrow & & \searrow \\ X & & Y, \end{array} \quad (1.4)$$

Since symmetric monoidal functors preserve traces by Lemma A.4, we get a natural equivalence

$$\text{tr}(f^*: \mathcal{D}(X) \rightarrow \mathcal{D}(X)) \simeq \mathcal{D}(\text{tr}(X \xleftarrow{\text{id}} X \xrightarrow{\text{id}} X)),$$

where the trace of $X \xleftarrow{\text{id}} X \xrightarrow{\text{id}} X$ is taken in the span category $\text{Span}(\mathcal{C})_{\mathcal{E}}$. Using the description of the evaluation and coevaluation on X from the discussion preceding the theorem, we then find that the trace of the morphism (1.4) in the span category is given by the outer span in the diagram:

$$\begin{array}{ccccc}
& & P' & & \\
& \swarrow & \downarrow \smile & \searrow & \\
& P & & X & \\
& \swarrow \quad \searrow & & \swarrow \Delta & \searrow \smile \\
X & & X \times X & & X \\
\swarrow \quad \searrow \Delta & \swarrow f \times \text{id} & \swarrow \smile & \searrow \Delta & \searrow \\
\text{pt} & X \times X & X \times X & X & \text{pt},
\end{array} \tag{1.5}$$

where by the pasting law the top corner P' can be computed as the pullback (1.2), i.e. there is a canonical equivalence $P' \simeq X^f$. Note also that X^f is separated since $X^f \rightarrow X$ is pulled back from $\Delta: X \rightarrow X \times X$ and thus admissible, whence the composition $X^f \rightarrow X \rightarrow \text{pt}$ is admissible.

If f is admissible, the equation (1.3) follows by dualizing (1.1), which interchanges $(-)^*$ and $(-)_!$. $\square$

Remark 1.4. Note that Proposition 1.3 looks like a utopian version of the Lefschetz trace formula from topology; it says that for any map $f \curvearrowright X$, the trace of the induced map on the categorified cohomology $\mathcal{D}(X)$ is isomorphic to the compactly-supported cohomology of the fixed points X^f . This lacks several subtleties that are present in the actual Lefschetz trace formula. For instance, the theorem neither has a transversality assumption on f nor local fixed point indices to correct for the lack of transversality, and there is no compactness/properness assumption on X . These subtleties arise from the failure of (1.1) to decategorify nicely.

Assume for simplicity that $\mathcal{A} = \text{Sp}$, as in the case of sheaves on locally compact Hausdorff spaces or local systems on anima. The naive approach to decategorification here is to apply topological Hochschild homology

$$\text{THH}: \text{Pr}_{\text{st}}^\diamond \rightarrow \text{Sp}$$

on both sides of the formula. On the left-hand side, one would then want to relate THH of $\text{tr}(f^* \curvearrowright \mathcal{D}(X))$ to the trace of map induced by f on $\text{THH}(\mathcal{D}(X)) \simeq \Gamma_c(X; \mathbb{S})$. If $\mathcal{D}(X)$ were dualizable *as an object of* $\text{Pr}_{\text{st}}^\diamond$, this would be formal, since THH is symmetric monoidal and thus preserves traces of endomorphisms of dualizable objects. However, for a dualizable category to be dualizable in dualizable categories (aka *fully dualizable* or *fully saturated*) is very restrictive. Indeed, [Har26] shows that the category of sheaves $\text{Shv}(X; \text{Sp})$ on a locally compact Hausdorff space X is fully dualizable if and only if X is a finite set of points. Similar subtleties surrounding decategorification and commuting THH with trace also play a central role in [BP24].

1.2 Naturality

It will be useful to know that (1.1) is a natural equivalence in the situations that we care about. For this to make sense, we must have a functoriality in mind for both the left and right sides of the equation. This functoriality is brought to you by 2-categorical enhancements of the six-functor formalism $\mathcal{D}$ à la [CLL26], as we now explain.

Let $(\mathcal{I}, \mathcal{P})$ be a suitable decomposition of $\mathcal{E}$ [Man22, Def A.5.9].

1.3 Decategorification

1.4 Examples

For each of the fixed point theorems contained in this article, the first step is applying [Proposition 1.3](#) to a concrete six-functor formalism.

Locally compact Hausdorff spaces

Anima

Sheaves of anima

2 The Lefschetz fixed point formula

In this section, we give a categorical proof of the following celebrated theorem of Lefschetz [[Lef26](#); [Lef37](#)]:

Theorem 2.1 (Lefschetz fixed point theorem). *Let $f: X \rightarrow X$ be a self-map of a compact ENR¹ and write $X^f \subseteq X$ for the fixed points of f . There exists a class $\chi_f \in H^0(X^f; \omega_{X^f})$ such that*

$$\langle 1, \chi_f \rangle = \sum_{i \geq 0} (-1)^i \text{tr}(H^i(f)) \sim H^i(X; \mathbb{Q}). \quad (2.1)$$

In particular, if the right-hand side is nonzero, then f has at least one fixed point.

If X is moreover a manifold, then the left-hand side can be identified with the intersection number of the graph of f with the diagonal in $X \times X$.

Note that the right-hand side of (2.1)—known as the *Lefschetz trace* of f —only depends on the homotopy class of f . The Lefschetz fixed point theorem is useful because it relates the space of fixed points to a homotopy invariant quantity, even though the space of fixed points can vary enormously between homotopic maps. In later sections we will consider a more refined variation on this theme, namely Lefschetz–Nielsen theory. We approach the Lefschetz fixed point formula and Lefschetz–Nielsen theory in the same way, namely by (1) proving a “formula” involving various categories of (co)sheaves, and then (2) decategorifying this to get statements about numbers and homotopy classes. Besides these steps, we also consider the problem of (3) identifying the numbers $\chi_f(a)$ appearing in (2.1).

We are by no means the first to use sheaf categories and trace functoriality to prove Lefschetz fixed point formulae. Indeed, our proof of the topological Lefschetz fixed point formula largely follows previous proofs in the algebro-geometric setting [[BN21](#); [KP20](#)]. The purpose of this section is simply to record this approach in the topological setting, and to serve as a warmup before the sections on Lefschetz–Nielsen theory below.

2.1 The Lefschetz trace

The first item on our agenda is to reinterpret the right-hand side of the Lefschetz fixed point formula (2.1) at the level of categories. For $h: Y \rightarrow Z$ some map of anima, we write $g^*: \mathbb{S}^Z \rightarrow \mathbb{S}^Y$ for the induced map on spherical cochains. Recall that if X is some compact anima, then by Spanier–Whitehead duality the spectrum $\mathbb{S}^X$ is dualizable, with dual given by the spherical chains $\mathbb{S}[X]$. We find it useful to take the following definition as our starting point:

¹More generally, a compact Hausdorff space that is locally of singular shape in the sense of [[Lur17](#), Definition A.4.15].

Define the evaluation pairing between Borel–Moore homology and compactly supported global sections.

Shorten this paragraph, e.g. by removing references to later sections.

Definition 2.2. For a compact anima X , the *Lefschetz trace* of a self-map $g: X \rightarrow X$ is the map

$$L(g) := \mathrm{tr}_{\mathrm{Sp}}(g^*: \mathbb{S}^X \rightarrow \mathbb{S}^X): \mathbb{S} \rightarrow \mathbb{S}.$$

If $X = \Pi_\infty(X)$ is the underlying anima of a topological space X and g is induced by a self-map $f: X \rightarrow X$, then we will also refer to $L(f) := L(g)$ as the Lefschetz trace of f .

Following [KP20], we will rewrite the Lefschetz trace using the machinery of higher traces provided by [HSS17] (see Section A for a brief overview). That is, let $\mathcal{P}_{\mathrm{st}}^{\mathrm{L}}$ denote the natural $(\infty, 2)$ -categorical enhancement of $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$. Morita theory provides a natural equivalence $\Omega \mathcal{P}_{\mathrm{st}}^{\mathrm{L}} \simeq \mathrm{Sp}$, whose inverse takes a spectrum $E \in \mathrm{Sp}$ to the endomorphism $E \otimes -: \mathrm{Sp} \rightarrow \mathrm{Sp}$. Moreover, under this equivalence, for any self map of a compact anima $g: X \rightarrow X$ the induced map of spectra $g^*: \mathbb{S}^X \rightarrow \mathbb{S}^X$ is identified with a map

$$g^*: (\mathrm{Sp}, \mathbb{S}^X) \rightarrow (\mathrm{Sp}, \mathbb{S}^X) \in \mathrm{End}^{\mathrm{tr}}(\mathcal{P}_{\mathrm{st}}^{\mathrm{L}})$$

We write

$$\mathrm{Tr}^{\mathrm{L}}: \mathrm{End}^{\mathrm{tr}}(\mathcal{P}_{\mathrm{st}}^{\mathrm{L}}) \longrightarrow \mathrm{Sp}.$$

for the symmetric monoidal trace functor of Theorem A.6. Unwinding the naturality of the functor Tr^{L} , we learn that $\mathrm{Tr}^{\mathrm{L}}(\mathrm{Sp}, \mathbb{S}^X) = \mathbb{S}$ and moreover:

$$L(g) = \mathrm{tr}_{\mathrm{Sp}}(g^*: \mathbb{S}^X \rightarrow \mathbb{S}^X) \simeq \mathrm{Tr}^{\mathrm{L}}(g^*: (\mathrm{Sp}, \mathbb{S}^X) \rightarrow (\mathrm{Sp}, \mathbb{S}^X)) \in \pi_0 \mathbb{S}. \quad (2.2)$$

With this in hand, our strategy for factoring the map $L(g)$ is to instead show that the map $g^*: (\mathrm{Sp}, \mathbb{S}^X) \rightarrow (\mathrm{Sp}, \mathbb{S}^X)$ in the category $\mathrm{End}^{\mathrm{tr}}(\mathcal{P}_{\mathrm{st}}^{\mathrm{L}})$.

2.2 Factoring the Lefschetz trace using sheaves

Above we showed that the Lefschetz trace of a self-map of a compact anima X can be calculated in terms of the lax commutative diagram

$$\begin{array}{ccc} \mathrm{Sp} & \xrightarrow{\mathbb{S}^X \otimes -} & \mathrm{Sp} \\ \parallel & \nearrow g^* & \parallel \\ \mathrm{Sp} & \xrightarrow{\mathbb{S}^X \otimes -} & \mathrm{Sp} \end{array}. \quad (2.3)$$

Suppose now that X is the underlying anima of a topological space X , and that $f: X \rightarrow X$ is some representative of the map g . We wish to express the trace of the diagram (2.3) in terms of the actual topological space A and the actual continuous map f . The basic idea is that sheaf theory accomplishes this under some mild assumptions on A .

Let Top denote the category of locally compact Hausdorff spaces. For any $X \in \mathrm{Top}$, we denote by

$$\mathcal{D}(X) := \mathrm{Shv}(X; \mathrm{Sp}) \in \mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$$

the ∞ -category of sheaves of spectra on X . By [Vol23], the pullback and compactly supported pushforward functors refine to a symmetric monoidal functor

$$\mathcal{D}_!^*(-): \mathrm{Span}(\mathrm{Top}) \longrightarrow \mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$$

which takes a span $B \xleftarrow{f} X \xrightarrow{g} C$ to the composite $\mathcal{D}(B) \xrightarrow{f^*} \mathcal{D}(X) \xrightarrow{g_!} \mathcal{D}(C)$. For any $X \in \mathrm{Top}$, we also write X^* and $X_!$ for the functors induced by the projection $X \rightarrow \mathrm{pt}$. Moreover, for some $\mathcal{F} \in \mathcal{D}(X)$ we write $\Gamma_c(X; \mathcal{F}) := X_! \mathcal{F}$ as well as $\Gamma_c(X) := \Gamma_c(X; \mathbb{S}_X) = X_! X^* \mathbb{S}$.

If X satisfies some mild conditions, the spectrum of spherical cochains on X is equivalent to the spherical “sheaf cohomology” of X , i.e. the global sections of the constant sheaf $\mathbb{S}_X$. This allows us to factor the horizontal functors in (2.3) as

$$\mathrm{Sp} \xrightarrow{A^*} \mathcal{D}(X) \xrightarrow{X_*} \mathrm{Sp}.$$

The following paragraph should maybe go in a notation and conventions section.

The category $\mathcal{D}(X)$ contains topological information about X .² We will use this to relate the Lefschetz trace to the fixed points $X^f \subseteq X$.

Proposition 2.3. *Let X be a compact Hausdorff space that is locally of singular shape, and let $f: X \rightarrow X$ be a self-map. Denote the underlying anima of X by X and the induced self-map on X by $g: X \rightarrow X$. Considered as a morphism in $\text{End}^{\text{tr}}(\text{Pr}_{\text{st}}^{\text{L}})$, the lax commutative square (2.3) factors as*

$$\left(\begin{array}{ccc} \mathcal{D}(X) & \xrightarrow{X_*} & \text{Sp} \\ \downarrow f_! & & \parallel \\ \mathcal{D}(X) & \xrightarrow{X_*} & \text{Sp} \end{array} \right) \circ \left(\begin{array}{ccc} \text{Sp} & \xrightarrow{X^*} & \mathcal{D}(X) \\ \parallel & \nearrow & \downarrow f_! \\ \text{Sp} & \xrightarrow{X^*} & \mathcal{D}(X) \end{array} \right), \quad (2.4)$$

where $X^* \rightarrow f_! f^* X^* \simeq f_! X^*$ comes from the unit transformation of the adjunction $f^* \dashv f_!$.

Proof. First note that both of the lax commutative squares appearing in (2.4) are morphisms in $\text{End}^{\text{tr}}(\text{Pr}_{\text{st}}^{\text{L}})$, or in other words that the horizontal functors are strongly continuous. Indeed, we have adjunctions

$$X^* \dashv X_* \simeq X_! \dashv X^!$$

where the equivalence uses that X is compact, and for the other square we find

$$X_* \simeq X_! \dashv X^! \simeq X^! \mathbb{S} \otimes X^*(-) \dashv X_* \mathcal{H}\text{om}(X^! \mathbb{S}, -),$$

where the equivalence $X^! \simeq X^! \mathbb{S} \otimes X^*$ uses that X is locally of singular shape, see [Vol23, Prop 6.18].

We must show that the lax commutative square (2.3) is equal to the outer lax commutative square in the diagram

$$\begin{array}{ccccc} \text{Sp} & \xrightarrow{X^*} & \mathcal{D}(X) & \xrightarrow{X_*} & \text{Sp} \\ \parallel & \nearrow & \downarrow f_! & & \parallel \\ \text{Sp} & \xrightarrow{X^*} & \mathcal{D}(X) & \xrightarrow{X_*} & \text{Sp}. \end{array} \quad (2.5)$$

This follows from [Lur17, Appendix A.4]. □

Using the functoriality of Tr^{L} , we thus get

Corollary 2.4. *Let X be a compact Hausdorff space that is locally of singular shape, and let $f: X \rightarrow X$ be some self-map. There is a commutative diagram*

$$\begin{array}{ccc} & \text{tr}(f_! \curvearrowright \mathcal{D}(X)) & \\ \nearrow & & \searrow \\ \mathbb{S} & \xrightarrow{L(f)} & \mathbb{S}, \end{array} \quad (2.6)$$

where the bottom arrow is the Lefschetz trace of f .

We now identify the top corner in the diagram (2.6). It will be useful to know that this identification is natural in some collection of maps. More precisely:

Definition 2.5. Given two pairs (X, f) and (X', f') consisting of a locally compact Hausdorff space and a self-map, a map $h: X \rightarrow X'$ is called *intertwining* if the diagram

$$\begin{array}{ccc} X & \xrightarrow{h} & X' \\ \downarrow f & & \downarrow f' \\ X & \xrightarrow{h} & X \end{array}$$

commutes.

²In fact, Aoki has shown that the homeomorphism type of a sober topological space can be reconstructed from its category of spectrum-valued sheaves together with the pointwise symmetric monoidal structure, see [Aok24].

Note that if $h: (X, f) \rightarrow (X', f')$ is intertwining and proper, then it gives rise to a lax commutative square

$$\begin{array}{ccc} \mathcal{D}(X) & \xrightarrow{h^*} & \mathcal{D}(X') \\ \downarrow f_! & \nearrow & \downarrow f'_! \\ \mathcal{D}(X) & \xrightarrow{h^*} & \mathcal{D}(X'), \end{array} \quad (2.7)$$

where the 2-cell is the Beck–Chevalley morphism. Together with the functor Tr^L , this lets us view the assignment $(X, f) \mapsto \mathrm{tr}(f_!: \mathcal{D}(X) \rightarrow \mathcal{D}(X))$ as a functor in the category of pairs (X, f) with proper morphisms between them.

Lemma 2.6. *Let $f: X \rightarrow X \in \mathrm{Top}$ be a self-map of a locally compact Hausdorff space. There is an equivalence of spectra*

$$\mathrm{tr}(f_!: \mathcal{D}(X) \rightarrow \mathcal{D}(X)) \simeq \Gamma_c(X^f),$$

where $X^f \subseteq X$ denotes the space of fixed points. Furthermore, this equivalence is natural in proper intertwining maps $h: (X, f) \rightarrow (X', f')$.

Proof. In terms of the symmetric monoidal functor $\mathcal{D}_!^*(-): \mathrm{Span}(\mathrm{Top}) \rightarrow \mathrm{Pr}_{\mathrm{st}}^L$, we can write

$$f_! \simeq \mathcal{D}_!^*(X \xleftarrow{\mathrm{id}} X \xrightarrow{f} X).$$

But symmetric monoidality then furnishes an equivalence

$$\mathrm{tr}(f_!: \mathcal{D}(X) \rightarrow \mathcal{D}(X)) \simeq \mathcal{D}_!^*(\mathrm{tr}(X \xleftarrow{\mathrm{id}} X \xrightarrow{f} X)),$$

where the trace of $X \xleftarrow{\mathrm{id}} X \xrightarrow{f} X$ is taken in the span category $\mathrm{Span}(\mathrm{Top})$. In the latter, the coevaluation and evaluation of the self-dual object X are given by

$$\begin{array}{ccc} & X & \\ \swarrow & & \searrow \Delta \\ \mathrm{pt} & & X \times X \end{array} \quad \text{and} \quad \begin{array}{ccc} & X & \\ \swarrow \Delta & & \searrow \\ X \times X & & \mathrm{pt}, \end{array}$$

respectively. The trace of $X \xleftarrow{\mathrm{id}} X \xrightarrow{f} X$ is therefore the outer span in the diagram

$$\begin{array}{ccccc} & & P' & & \\ & & \vee & & \\ & X & & P & \\ & \swarrow \quad \searrow & & \swarrow \quad \searrow & \\ & X & X \times X & X & \\ & \swarrow \quad \searrow & \swarrow \quad \searrow & \swarrow \quad \searrow & \\ \mathrm{pt} & X \times X & X \times X & X \times X & \mathrm{pt}, \end{array} \quad (2.8)$$

where by the pasting law the top corner can be computed as the pullback of the cospan

$$X \xrightarrow{(f, \mathrm{id})} X \times X \xleftarrow{\Delta} X,$$

Find the right things to cite to make this into a functor from category of N-equivariant spaces with proper morphisms between them. It's probably in Lurie that going from commutative squares (with vertical left adjoints) to lax commutative squares by vertically adjoining and using Beck–Chevalley 2-cell is a functor.

and this is simply the space of fixed points X^f .

As for naturality, assume that $h: (X, f) \rightarrow (X', f')$ is a proper map. Then the induced map

$$\mathrm{tr}(f_! : \mathcal{D}(X) \rightarrow \mathcal{D}(X)) \rightarrow \mathrm{tr}(f'_! : \mathcal{D}(X') \rightarrow \mathcal{D}(X')),$$

is given in $\Omega\mathcal{P}\mathrm{r}_{\mathrm{st}}^{\mathrm{L}}$ by the composition of 2-cells

$$\begin{array}{ccccccc} \mathrm{Sp} & \xrightarrow{\mathrm{coev}} & \mathcal{D}(X') \otimes \mathcal{D}(X') & \xrightarrow{f'_! \otimes \mathrm{id}} & \mathcal{D}(X') \otimes \mathcal{D}(X') & \xrightarrow{\mathrm{ev}} & \mathrm{Sp} \\ \parallel & \swarrow & \downarrow h^* \otimes h^* & \swarrow \mathrm{BC} \otimes \mathrm{id} & \downarrow h^* \otimes h^* & \swarrow & \parallel \\ \mathrm{Sp} & \xrightarrow{\mathrm{coev}} & \mathcal{D}(X) \otimes \mathcal{D}(X) & \xrightarrow{f_! \otimes \mathrm{id}} & \mathcal{D}(X) \otimes \mathcal{D}(X) & \xrightarrow{\mathrm{ev}} & \mathrm{Sp}, \end{array}$$

which is induced by the composition of 2-cells in $\mathrm{Span}(\mathrm{Top})$

$$\begin{array}{ccccccc} & & X & & X \times X & & X \\ & \swarrow & \Delta & & \swarrow & \searrow & \swarrow \\ \mathrm{pt} & & h & & X \times X & & X \\ & \swarrow & \Delta & & \swarrow & \searrow & \swarrow \\ & & X' & & X' \times X' & & X' \\ & \swarrow & \Delta & & \swarrow & \searrow & \swarrow \\ \mathrm{pt} & & X' \times X' & & X' \times X' & & \mathrm{pt}, \end{array}$$

and taking pullbacks to compose these morphisms in the span category shows that the 2-cell identifies with

$$\begin{array}{ccccc} & & X^f & & \\ & \swarrow & \Delta & \searrow & \\ \mathrm{pt} & & (X')^{f'} & & \mathrm{pt} \\ & \swarrow & \Delta & \searrow & \\ \mathrm{pt} & & & & \mathrm{pt}, \end{array}$$

where the middle vertical arrow is the restriction of h to X^f . □

Combining this with [Corollary 2.4](#), we deduce:

Theorem 2.7 (Lefschetz fixed point formula). *Let X be a compact Hausdorff space that is locally of singular shape, and let $f: X \rightarrow X$ be a self-map. There is a commutative diagram*

$$\begin{array}{ccc} & \Gamma_c(X^f) & \\ \nearrow & & \searrow \\ \mathbb{S} & \xrightarrow{L(f)} & \mathbb{S}, \end{array} \tag{2.9}$$

where the bottom arrow is the Lefschetz trace of f and the arrow $\mathbb{S} \rightarrow \Gamma_c(X^f)$ is the one induced by the projection $X^f \rightarrow \mathrm{pt}$, or in other words the unit in the sheaf cohomology ring $\Gamma_c(X^f) \simeq \Gamma(X^f)$.

Proof. It remains only to justify the statement about the map $\mathbb{S} \rightarrow \Gamma_c(X^f)$. Recall that (2.6) is induced by the factorization (2.4) of the lax commutative square (2.3). In this factorization, the first factor is precisely the lax square induced by the proper morphism $(X, f) \rightarrow (\text{pt}, \text{id})$ as in (3.3). The desired conclusion now follows from the naturality statement in Lemma 2.6. $\square$

We immediately get:

Corollary 2.8 (Lefschetz fixed point theorem). *Let A and f be as in the previous theorem. If the Lefschetz trace of f is nonzero, then f has a fixed point.*

Proof. If X^f is empty, then $\Gamma_c(X^f) \simeq 0$ and so the Lefschetz trace must be null. $\square$

Definition 2.9. Let $X \in \text{Top}$ be compact and locally of singular shape. For any self map $f: X \rightarrow X$, we call the map

$$\lambda_f: \Gamma_c(X^f) \rightarrow \mathbb{S},$$

appearing as the right hand diagonal arrow in Equation (2.9), the *topological Reidemeister trace* of f .

Finally, we show how to recover Theorem 2.1 from Theorem 2.7.

Proof of Theorem 2.1. First, note that for any compact anima X and a self-map $g: X \rightarrow X$ we have that:

$$\sum_{i \geq 0} (-1)^i \text{tr}((H^i(g) \curvearrowright H^i(X; \mathbb{Q}))) = L(g).$$

Indeed, this is immediate since both the rationalization functor $\text{Sp} \rightarrow \text{Mod}_{\mathbb{Q}}$ and the homotopy group functor $\pi_*: \text{Mod}_{\mathbb{Q}} \rightarrow \text{grVect}_{\mathbb{Q}}$ are symmetric monoidal.

Base changing (2.6) along the Hurewicz map $\mathbb{S} \rightarrow \mathbb{Q}$ and using the projection formula $\mathbb{Q} \otimes \Gamma_c(X^f) \simeq \Gamma_c(X^f; \mathbb{Q})$, we get a commutative diagram

$$\begin{array}{ccc} & \Gamma_c(X^f; \mathbb{Q}) & \\ \nearrow & & \searrow \\ \mathbb{Q} & \xrightarrow{L(f)} & \mathbb{Q} \end{array}$$

in the derived category of $\mathbb{Q}$ -modules, where the bottom horizontal map is the Lefschetz trace and the map $\mathbb{Q} \rightarrow \Gamma_c(X^f; \mathbb{Q}) \simeq \Gamma(X^f; \mathbb{Q})$ is the unit of the sheaf cohomology ring. Here

$$\pi_0 \text{Map}(\Gamma_c(X^f; \mathbb{Q}), \mathbb{Q}) \simeq H_0(X^f_! (X^f)^! \mathbb{Q}),$$

and the right-hand side is the zeroth sheaf homology of X^f . $\square$

Finish

2.3 Local indices

Say that $X \in \text{Top}$ is *suave* if the map $X \rightarrow \text{pt}$ is a shape submersion in the sense of [Vol23]. In what follows, we fix a compact suave topological space $X \in \text{Top}$ together with a self map $f: X \rightarrow X$.

Let us now analyze the other maps appearing in the diagram above. The map $\text{Tr}^L(X^*, \sigma)$ is given by the composite natural transformation

$$\mathbb{S} = \text{tr}_L(\text{id}_{\text{Sp}}) \rightarrow \text{tr}_L(A_* A^*) \simeq \text{tr}_L(A^* A_*) \xrightarrow{\sigma} \text{tr}_L(f_! A^* A_*) \rightarrow \text{tr}_L(f_!).$$

Dually, the map $\text{Tr}^L(X_!, \sigma)$ is given by the composite

$$\text{tr}_L(f_!) \rightarrow \text{tr}_L(A^! A_! f_!) \xrightarrow{\text{can}} \text{tr}_L(A^! A_!) \simeq \text{tr}_L(A_! A^!) \rightarrow \text{tr}_L(\text{id}_{\text{Sp}}) = \mathbb{S}.$$

In particular, we obtain a further factorization of the trace as

This subsection still needs to be edited to conform with the changes in preceding subsections, such as the fact that we now already have an identification of the leftmost leg in the Russian triangle.

$$\mathbb{S} \rightarrow \Gamma(A) \xrightarrow{a_f} \mathrm{tr}_L(f!) \xrightarrow{b_f} \Gamma(A; \omega_A) \rightarrow \mathbb{S},$$

where the left and rightmost maps are the unit and counit maps respectively. We give an explicit construction of these maps following that of Kashiwara–Shapira [KS90].

Construction 2.10. Write $j: X^f \rightarrow X$ for the inclusion of the fixed points, $\Delta: X \rightarrow X \times X$ for the diagonal, $\pi_i: X \times X \rightarrow X$ for the projections and $\Gamma^f := (f, \mathrm{id}): X \rightarrow X \times X$ for the graph of f . Combining the equivalences

$$\mathbb{S}_X \simeq \Delta^!(\pi_1^* \mathbb{S}_X \otimes \pi_2^* \omega_X) \simeq \Delta^! \pi_2^* \omega_X$$

with the unit transformation $\mathrm{id} \rightarrow \Gamma_*^f(\Gamma^f)^*$ and the Beck–Chevalley equivalence $\Delta^! \Gamma_*^f \simeq j_* j^!$ we obtain a map of sheaves

$$\mathbb{S}_X \longrightarrow \Delta^! \Gamma_*^f(\Gamma^f)^* \pi_2^* \omega_X \simeq \Delta^! \Gamma_*^f \omega_X \simeq j_* j^! \omega_X.$$

Further composing with the counit of the adjunction $j_! \dashv j^!$ we obtain a diagram of sheaves

$$\mathbb{S}_A \rightarrow j_* j^! \omega_A \rightarrow \omega_A.$$

Taking the dual $\mathbb{D}(-) = \mathcal{H}om_{\mathcal{D}(A)}(-, \omega_A)$ and using that $\mathbb{D} \circ j_! = j_* \circ \mathbb{D}$ by Lemma B.6 as well as $\mathbb{D}(\omega_A) = \mathbb{S}_A$ since A is suave, we get a further diagram of sheaves

$$\mathbb{S}_A \xrightarrow{\alpha_f} j_* j^* \mathbb{S}_A \xrightarrow{\beta_f} \omega_A.$$

Note that, α_f is simply induced by the unit natural transformation $\mathrm{id}_A \rightarrow j_* j^*$.

Lemma 2.11. *For any self map $f: X \rightarrow X$ of a compact suave space $X \in \mathrm{Top}$, we have natural equivalences $\Gamma(\alpha_f) = a_f$ as well as $\Gamma(\beta_f) = b_f$.*

Do proof

Note that, since X is compact suave, the closed subspace $X^f \subseteq X$ is as well, so we have a natural identification $\Gamma(X^f)^\vee \simeq \Gamma(X^f; \omega_X)$. Thus, we may think of χ_f as a class in the homology $\lambda_f \in H_0(X; \mathbb{S}) = H_0(X; \mathbb{Z})$, i.e. a formal sum

$$\lambda_f = \sum_{[a] \in \pi_0 X^f} \lambda_f^a [a] \in \mathbb{Z}[\pi_0 X^f].$$

For any $a \in X^f$, we call $\lambda_f^a \in \mathbb{Z}$ the *Lefschetz index* of f at a .

Lemma 2.12. *If X is a homology manifold, meaning we have a trivialization $\omega_X \simeq X^* \mathbb{S}[n]$, the sum of the indices is given by the intersection number of the graph with the diagonal*

$$\sum_{[a] \in \pi_0 X^f} \lambda_f^a = \#(\Gamma^f \cap \Delta).$$

Proof. Cite KS

□

3 Lefschetz–Nielsen theory

3.1 Local systems and the Reidemeister trace

Let $g: X \rightarrow X$ be a self-map of a compact anima X . Consider again the lax commutative square (2.3) from Section 2, which we viewed as a morphism in the category of traceable endomorphisms $\mathrm{End}^{\mathrm{tr}}(\mathcal{P}_{\mathrm{st}}^L)$. If $g: X \rightarrow X$ happened to be induced by a self-map $f: X \rightarrow X$ for some well-behaved space X , we recovered the Lefschetz fixed point theorem for f by factoring (2.3) through the object $(\mathcal{D}(X), f_!) \in \mathrm{End}^{\mathrm{tr}}(\mathcal{P}_{\mathrm{st}}^L)$.

Putting topology aside for now, a more obvious factorization of (2.3) is through the category of local systems (aka parametrized spectra) Sp^X :

Another thing that might belong in this section is something like $\chi_{f \times f'} = \chi_f \times \chi_{f'}$, just as a demonstration of how easy it is to prove things like this using the categorical framework

Proposition 3.1. *Let $g: X \rightarrow X$ be a self-map of a compact anima X . Considered as a morphism in $\text{End}^{\text{tr}}(\mathcal{P}\text{r}_{\text{st}}^{\text{L}})$, the lax commutative square (2.3) factors as*

$$\left(\begin{array}{ccc} \text{Sp}^X & \xrightarrow{X_{\sharp}} & \text{Sp} \\ \downarrow g^* & \nearrow & \parallel \\ \text{Sp}^X & \xrightarrow{X_{\sharp}} & \text{Sp} \end{array} \right) \circ \left(\begin{array}{ccc} \text{Sp} & \xrightarrow{X^*} & \text{Sp}^X \\ \parallel & & \downarrow g^* \\ \text{Sp} & \xrightarrow{X^*} & \text{Sp}^X \end{array} \right), \quad (3.1)$$

where the rightmost square 2-cell is the invertible one coming from functoriality of pullback, and the leftmost 2-cell is the counit transformation $X_{\sharp}g^* \simeq X_{\sharp}g_{\sharp}g^* \rightarrow X_{\sharp}$.

Proof. Aside from the usual adjunctions $X_{\sharp} \dashv X^* \dashv X_*$, the fact that X is compact furnishes an additional adjunction $X_* \dashv \text{Hom}_{\text{Sp}^X}(D_X, X^*(-))$, where D_X is the dualizing object of X . It follows that the horizontal morphisms in both squares are strongly continuous, and hence that they define morphisms in $\text{End}^{\text{tr}}(\mathcal{P}\text{r}_{\text{st}}^{\text{L}})$. Concatenating the two 2-cells recovers the one in (2.3) on the nose. $\square$

As in the previous section, we then get

Corollary 3.2. *For a self-map $g: X \rightarrow X$ of a compact anima X , there is a commutative diagram*

$$\begin{array}{ccc} & \text{tr}(g^* \curvearrowright \text{Sp}^X) & \\ \nearrow & & \searrow \\ \mathbb{S} & \xrightarrow{\quad} & \mathbb{S}, \end{array} \quad (3.2)$$

where the bottom arrow is the Lefschetz trace of g .

Just as in the previous section, the axioms of six-functor formalisms allow us to quickly compute the trace $\text{tr}(g^* \curvearrowright \text{Sp}^X)$ with the suspension spectrum of a space of “fixed points” for the map g . Here fixed points must be interpreted in a homotopy-invariant way; we let $\mathcal{L}^g X$ denote the homotopical fixed point space defined to be the homotopy pullback

$$\begin{array}{ccc} \mathcal{L}^g X & \xrightarrow{\quad} & X \\ \downarrow & \lrcorner & \downarrow \Delta \\ X & \xrightarrow{\Gamma^g} & X \times X. \end{array}$$

That is, we have

Lemma 3.3. *Let $g: X \rightarrow X$ be a self-map of a compact anima. There is an equivalence of spectra*

$$\text{tr}(g^* \curvearrowright \text{Sp}^X) \simeq \mathbb{S}[\mathcal{L}^g X],$$

natural in intertwining maps $h: (X, g) \rightarrow (X', g')$.

Proof. The identification comes about exactly as in the proof of Lemma 2.6. Unlike in the topological setting, we do not need to require anything of h beyond being an intertwining map, since any intertwining map induces a morphism

$$\begin{array}{ccc} \text{Sp}^X & \xrightarrow{h_{\sharp}} & \text{Sp}^{X'} \\ \downarrow g^* & \nearrow & \downarrow (g')^* \\ \text{Sp}^X & \xrightarrow{h_{\sharp}} & \text{Sp}^{X'} \end{array} \quad (3.3)$$

in $\text{End}^{\text{tr}}(\mathcal{P}\text{r}_{\text{st}}^{\text{L}})$, where the 2-cell is the Beck–Chevalley morphism. $\square$

Combining [Lemma 3.3](#) with [Corollary 3.2](#), we find

Theorem 3.4. *Let $g: X \rightarrow X$ be a self-map of a compact anima. There is a commutative triangle*

$$\begin{array}{ccc} & \mathbb{S}[\mathcal{L}^g X] & \\ \nearrow & & \searrow \\ \mathbb{S} & \xrightarrow{\quad} & \mathbb{S}, \end{array} \quad (3.4)$$

in which the bottom arrow is the Lefschetz trace and the arrow $\mathbb{S}[\mathcal{L}^g X] \rightarrow \mathbb{S}$ is induced by the projection $\mathcal{L}^g X \rightarrow \text{pt}$.

Proof. The second statement follows from the naturality of the identification in [Lemma 3.3](#). $\square$

We have constructed a homotopy invariant for self-maps that refines the Lefschetz trace:

Definition 3.5. The map $\rho_g: \mathbb{S} \rightarrow \mathbb{S}[\mathcal{L}^g X]$ in (3.4) is the *Reidemeister trace* of g .

3.2 Factoring the Reidemeister trace using cosheaves

Let $X \in \text{Top}$ be compact and locally of singular shape with associated anima $\Pi_\infty(X)X$. Given some self map $f: X \rightarrow X$, we can now associated two different invariants to this datum, namely the topological Reidemeister trace

$$\lambda_f: \mathbb{S} \rightarrow \Gamma(X; \omega_X),$$

as well as its homotopical counterpart:

$$\rho_f: \mathbb{S} \rightarrow \mathbb{S}[\mathcal{L}^f \Pi_\infty(X)].$$

The insight is that the Lefschetz–Nielsen theorem, [will be an immediate consequence of relating](#) these two invariants. More precisely, we show:

Add ref

Theorem 3.6. *Let $X \in \text{Top}$ be compact and locally of singular shape. Then we have a commutative diagram in the category of spectra:*

$$\begin{array}{ccc} & \Gamma(X, \omega_X) & \\ \lambda_f \nearrow & & \downarrow \alpha \\ \mathbb{S} & \xrightarrow[\rho_f]{} & \mathbb{S}[\mathcal{L}^f \Pi_\infty(X)] \end{array}$$

which is natural in intertwining maps.

In the spirit of fixed point theory, we would like to relate the Reidemeister trace of g to the actual fixed point space X^f . We do so by following the pattern of [Section 2](#). First, then, we should factor the square

$$\begin{array}{ccc} \text{Sp} & \xrightarrow{X^*} & \text{Sp}^X \\ \parallel & & \downarrow g^* \\ \text{Sp} & \xrightarrow{X^*} & \text{Sp}^X \end{array} \quad (3.5)$$

through an object in $\text{End}^{\text{tr}}(\mathcal{P}\text{r}_{\text{st}}^{\text{L}})$ that has access to actual topological information about the pair (A, f) . Such a factorization is achieved by the *assembly functor* of Bartels and Nikolaus. Although this work is the subject of a still unpublished article, an expository account can be found in the lecture notes [\[Nik24\]](#).

Definition 3.7 (Bartels–Nikolaus). Let $(\mathcal{C}, \otimes, 1)$ be a closed symmetric monoidal category with internal mapping object $\mathcal{H}om_{\mathcal{C}}(-, -)$. Given objects A and B in $\mathcal{C}$ together with a map

$$\delta: 1 \rightarrow A \otimes B,$$

the associated δ -assembly map, denoted a_{δ} , is defined by the following string diagram

$$a_{\delta} = \begin{array}{c} B^{\vee} \\ \downarrow \\ \text{ev}_B \\ \uparrow \\ B \\ \delta \\ \downarrow \\ A, \end{array}$$

where $B^{\vee} = \mathcal{H}om_{\mathcal{C}}(B, 1)$ and $\text{ev}_B: B^{\vee} \otimes B \rightarrow 1$ is the evaluation map, i.e. the counit of the adjunction between $- \otimes B$ and $\mathcal{H}om_{\mathcal{C}}(B, -)$.

Let $\text{Pr}_{\text{st}}^{\diamond}$ denote the category of dualizable stable categories with strongly continuous functors between them. The Lurie tensor product on presentable categories restricts to a symmetric monoidal structure on $\text{Pr}_{\text{st}}^{\diamond}$. By [Ram24], this symmetric monoidal structure is closed. We denote the resulting internal mapping object by $\mathcal{H}om^{\diamond}(-, -)$.

Definition 3.8 (Bartels–Nikolaus). Let X be a locally compact Hausdorff space. The category of *completed cosheaves* on X , denoted $\check{\mathfrak{D}}(X)$, is given by

$$\check{\mathfrak{D}}(X) = \mathcal{H}om^{\diamond}(\mathcal{D}(X), \text{Sp}).$$

Although general objects of $\check{\mathfrak{D}}(X)$ are quite mysterious, the definition does give us a description of the compact objects via

$$\check{\mathfrak{D}}(X)^{\omega} \simeq \text{Map}_{\text{Pr}_{\text{st}}^{\diamond}}(\text{Sp}, \check{\mathfrak{D}}(X)) \simeq \text{Map}_{\text{Pr}_{\text{st}}^{\diamond}}(\text{Sp}, \mathcal{H}om^{\diamond}(\mathcal{D}(X), \text{Sp})) \simeq \text{Map}_{\text{Pr}_{\text{st}}^{\diamond}}(\mathcal{D}(X), \text{Sp}),$$

where the final equivalence used the adjunction between internal mapping object and $\otimes$. That is, the groupoid of compact objects $\check{\mathfrak{D}}(X)^{\omega}$ is equivalent to the groupoid strongly continuous functors $\mathcal{D}(X) \rightarrow \text{Sp}$, which is a subgroupoid of the more familiar groupoid of all colimit-preserving functors $\mathcal{D}(X) \rightarrow \text{Sp}$. Of particular interest is the strongly continuous functor

$$X_{\sharp}: \mathcal{D}(X) \rightarrow \text{Sp}$$

that is left adjoint to X^* , defined if X is locally of singular shape. The corresponding compact object of $\check{\mathfrak{D}}(X)$ is denoted w_X and referred to as the *local Wall object* of X .

Another aspect of $\check{\mathfrak{D}}(X)$ which is well-understood by hard work of Bartels–Nikolaus and Efimov is its value under localizing invariants:

Theorem 3.9 (Bartels–Efimov–Nikolaus). $K(\check{\mathfrak{D}}(X)) \simeq X_* X^! K(\mathbb{S})$.

Finally, we will need to introduce some notation for the functoriality of $\check{\mathfrak{D}}(X)$. If $h: X \rightarrow X'$ is a proper map of locally compact Hausdorff spaces, we get as usual that the pullback $h^*: \mathcal{D}(X') \rightarrow \mathcal{D}(X)$ is strongly continuous. Hence it induces a map on completed cosheaves that we denote

$$h_{\sharp}: \check{\mathfrak{D}}(X) \rightarrow \check{\mathfrak{D}}(X').$$

That is, the category of completed cosheaves is *covariantly functorial in proper maps*. Since $h_{\sharp}$ is strongly continuous by construction, it admits a colimit-preserving left adjoint that we denote

$$h^{\sharp}: \check{\mathfrak{D}}(X') \rightarrow \check{\mathfrak{D}}(X).$$

The reader may observe that there is an additional contravariant functoriality in etale maps, for instance, that one would might want to denote h^ι .

Assume that X is locally of singular shape and compact. Under these assumptions, we have a sequence of colimit-preserving functors

$$\mathcal{D}(X) \otimes \mathrm{Sp}^{\Pi_\infty(X)} \xrightarrow{\mathrm{id} \otimes \psi^*} \mathcal{D}(X) \otimes \mathcal{D}(X) \xrightarrow{\Delta^*} \mathcal{D}(X) \xrightarrow{X^*} \mathrm{Sp},$$

where $\psi^*: \mathrm{Sp}^{\Pi_\infty(X)} \rightarrow \mathcal{D}(X)$ denotes the inclusion of local systems into all sheaves. Bartels and Nikolaus show that the composite $\mathcal{D}(X) \otimes \mathrm{Sp}^{\Pi_\infty(X)} \rightarrow \mathrm{Sp}$ admits a left adjoint

$$\delta: \mathrm{Sp} \rightarrow \mathcal{D}(X) \otimes \mathrm{Sp}^{\Pi_\infty(X)},$$

which is then a strongly continuous functor and thus a morphism in the category $\mathrm{Pr}_{\mathrm{st}}^\diamond$. Specializing Definition 3.7 to this situation, we thus get a strongly continuous functor

$$a_\delta: \check{\mathcal{D}}(X) \rightarrow \mathrm{Sp}^{\Pi_\infty(X)}, \quad (3.6)$$

the *Bartels–Nikolaus assembly functor*.

Theorem 3.10 (Bartels–Nikolaus). *Let X be a compact Hausdorff space that is locally of constant shape. Then the functor (3.6)*

- (i) *induces the Waldhausen assembly map on K -theory and*
- (ii) *maps the local Wall object w_X to the constant sphere spectrum $\mathbb{S}_{\Pi_\infty X}$.*

Part (ii) means that we can factor $(\Pi_\infty X)^*: \mathrm{Sp} \rightarrow \mathrm{Sp}^{\Pi_\infty X}$ through the Bartels–Nikolaus assembly functor. Better, we have

Proposition 3.11. *Let $f: X \rightarrow X$ be a self-map of a compact Hausdorff space that is locally of singular shape. Considered as a morphism in $\mathrm{End}^{\mathrm{tr}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$, the lax commutative square (3.5) factors as*

$$\left(\begin{array}{ccc} \check{\mathcal{D}}(X) & \xrightarrow{a_\delta} & \mathrm{Sp}^{\Pi_\infty(X)} \\ \downarrow f? & \nearrow & \downarrow f^* \\ \check{\mathcal{D}}(X) & \xrightarrow{a_\delta} & \mathrm{Sp}^{\Pi_\infty(X)} \end{array} \right) \circ \left(\begin{array}{ccc} \mathrm{Sp} & \xrightarrow{w_X} & \check{\mathcal{D}}(X) \\ \parallel & \nearrow & \downarrow f? \\ \mathrm{Sp} & \xrightarrow{w_X} & \check{\mathcal{D}}(X) \end{array} \right), \quad (3.7)$$

in which the leftmost 2-cell is the transpose of the commutative diagram

$$\begin{array}{ccc} \check{\mathcal{D}}(X) & \xrightarrow{a_\delta} & \mathrm{Sp}^{\Pi_\infty(X)} \\ \downarrow f? & & \downarrow f! \\ \check{\mathcal{D}}(X) & \xrightarrow{a_\delta} & \mathrm{Sp}^{\Pi_\infty(X)} \end{array},$$

and the rightmost 2-cell is the transpose of the 2-cell

$$\begin{array}{ccc} \mathrm{Sp} & \xrightarrow{w_X} & \check{\mathcal{D}}(X) \\ \parallel & \nearrow & \downarrow f? \\ \mathrm{Sp} & \xrightarrow{w_X} & \check{\mathcal{D}}(X) \end{array}$$

coming from the counit transformation

$$f?(w_X) = X_\# \circ f^* \simeq X_\# \circ f_\# \circ f^* \rightarrow X_\# = w_X.$$

Corollary 3.12. *Let $f: X \rightarrow X$ be a self-map of a compact Hausdorff space that is locally of singular shape. There is a commutative triangle*

$$\begin{array}{ccc} & \mathrm{tr}(f^? \curvearrowright \check{\mathfrak{D}}(X)) & \\ \nearrow & & \searrow \\ \mathbb{S} & \xrightarrow{\quad\quad\quad} & \mathbb{S}[\mathcal{L}^f \Pi_\infty X], \end{array} \quad (3.8)$$

where the bottom horizontal arrow is the Reidemeister trace.

3.3 Another trace “calculation”

Keeping with the pattern laid out in [Section 2](#), the next step in our treatment of Lefschetz–Nielsen theory is a calculation of the categorical trace that shows up as the apex in the triangle (3.8). Unlike in the Lefschetz and Reidemeister trace incarnations of this step, we will not be able to calculate $\mathrm{tr}(f^? \curvearrowright \check{\mathfrak{D}}(X))$ in the strict sense (without a big effort); instead, we will give a lax calculation that suffices for the purposes of fixed point theory. This idea is again adapted to our setting from unpublished work of Bartels and Nikolaus, who show using simple manipulations in closed monoidal categories that the map induced on K -theory by their assembly functor $a_\delta: \check{\mathfrak{D}}(X) \rightarrow \mathrm{Sp}^{\Pi_\infty(X)}$ factors through $X_* X^! K\mathbb{S}$, thereby removing the difficult [Theorem 3.9](#) from the dependencies of the customer-facing [Theorem 3.10](#).

We should expect the calculation of $\mathrm{tr}(f^? \curvearrowright \check{\mathfrak{D}}(X))$ to be difficult, because for $f = \mathrm{id}$ it is already a difficult theorem of Efimov.

Definition 3.13 (The free algebra on one dualizable object). $\mathrm{Sp}[T, T^\vee] = \mathrm{Fun}(\mathrm{Bord}_{\mathrm{fr}}^1(\mathbb{R}^1)^{\mathrm{op}}, \mathrm{Sp})$

motivate
linearizing
over free
algebra on
dualizable
generator

4 Parametrized Lefschetz–Nielsen theory

A Trace theory

Throughout this section, fix a symmetric monoidal $(\infty, 2)$ -category $(\mathcal{C}, \otimes)$ with monoidal unit $\mathbb{1}_{\mathcal{C}} \in \mathcal{C}$ and write $\Omega\mathcal{C}$ for the $(\infty, 1)$ -category of endomorphisms $\mathbb{1}_{\mathcal{C}} \rightarrow \mathbb{1}_{\mathcal{C}}$. The symmetric monoidal structure on $\mathcal{C}$ induces a natural symmetric monoidal structure on $\Omega\mathcal{C}$, such that the tensor product $f \otimes_{\Omega\mathcal{C}} g$ of two endomorphisms $f, g: \mathbb{1}_{\mathcal{C}} \rightarrow \mathbb{1}_{\mathcal{C}}$ is simply given by the map $f \otimes g: \mathbb{1}_{\mathcal{C}} \rightarrow \mathbb{1}_{\mathcal{C}}$.

Definition A.1. An object $X \in \mathcal{C}$ is said to be *dualizable* if the functor $X \otimes -: \mathcal{C} \rightarrow \mathcal{C}$ admits an adjoint of the form $Y \otimes -: \mathcal{C} \rightarrow \mathcal{C}$. In this case, we write $Y = X^\vee$ and refer to $X^\vee$ as the *dual* of X . We denote by $\mathcal{C}^\diamond \subseteq \mathcal{C}$ the full subcategory spanned by the dualizable objects.

Remark A.2. For any $X \in \mathcal{C}^\diamond$ we have a canonical equivalence $(X^\vee)^\vee \simeq X$ and the assignment $X \mapsto X^\vee$ refines to an equivalence of categories $\mathcal{C}^{\diamond, \mathrm{op}} \xrightarrow{\sim} \mathcal{C}^\diamond$.

By definition, any $X \in \mathcal{C}^\diamond$ comes equipped with evaluation and coevaluation maps

$$\mathrm{coev}_X: \mathbb{1}_{\mathcal{C}} \longrightarrow X \otimes X^\vee$$

$$\mathrm{ev}_X: X^\vee \otimes X \longrightarrow \mathbb{1}_{\mathcal{C}}$$

Moreover, since $\mathcal{C}$ is symmetric monoidal, there is a canonical *braiding*, giving a natural equivalence

$$\tau: X \otimes X^\vee \simeq X^\vee \otimes X.$$

Definition A.3. A *generalized endomorphism* of an object $X \in \mathcal{C}$ is a map

$$f: Y \otimes X \rightarrow X \otimes Z,$$

define Hopf
algebra
structure

for some $Y, Z \in \mathcal{C}$. If $X \in \mathcal{C}^\diamond$, the *trace* of f is defined to be the composite

$$\mathrm{tr}_{\mathcal{C}}(f): Y \xrightarrow{Y \otimes \mathrm{coev}_X} Y \otimes X \otimes X^\vee \xrightarrow{f \otimes \mathrm{id}} Z \otimes X \otimes X^\vee \xrightarrow{Z \otimes \tau} Z \otimes X^\vee \otimes X \xrightarrow{Z \otimes \mathrm{ev}_X} Z.$$

We refer to $\mathrm{tr}_{\mathcal{C}}(\mathrm{id}_X): \mathbb{1}_{\mathcal{C}} \rightarrow \mathbb{1}_{\mathcal{C}}$ as the *dimension* of X and denote it by $\dim_{\mathcal{C}}(X)$.

Lemma A.4. *Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a symmetric monoidal $(\infty, 2)$ -functor and let $f: Y \otimes X \rightarrow X \otimes Z$ be a generalized endomorphism of some dualizable object $X \in \mathcal{C}^\diamond$. We have a natural equivalence*

$$F(\mathrm{tr}_{\mathcal{C}}(f)) \simeq \mathrm{tr}_{\mathcal{D}}(F(f)),$$

in the $(\infty, 1)$ category of maps $F(Y) \rightarrow F(Z)$.

In [HSS17] Hoyois–Scherotzke–Sibilla construct an $(\infty, 1)$ -category $\mathrm{End}^{\mathrm{tr}}(\mathcal{C})$ of *traceable endomorphisms* in $\mathcal{C}$ as follows:

Construction A.5. The assignment $\mathcal{C} \mapsto \mathcal{C}^\diamond$ refines to a functor

$$(-)^\diamond: \mathrm{Cat}_{(\infty, 2)}^\otimes \longrightarrow \mathrm{Cat}_{(\infty, 2)}$$

which preserves filtered colimits and all limits by [Lur17, Proposition 4.6.1.11] and thus admits a left adjoint denoted by

$$\mathrm{Fr}^\diamond: \mathrm{Cat}_{(\infty, 2)} \longrightarrow \mathrm{Cat}_{(\infty, 2)}^\otimes.$$

The $(\infty, 1)$ -category of traceable endomorphisms $\mathrm{End}^{\mathrm{tr}}(\mathcal{C})$ is defined as the maximal sub- $(\infty, 1)$ -category of the $(\infty, 2)$ -category $\mathrm{Fun}_\otimes^{\mathrm{oplax}}(\mathrm{Fr}^\diamond(B\mathbb{N}), \mathcal{C})$ of symmetric monoidal oplax functors from the “dualizably free” $(\infty, 2)$ -category on the walking category with one endomorphism.

Explicitly, objects of $\mathrm{End}^{\mathrm{tr}}(\mathcal{C})$ are given by self maps of dualizable objects $f: X \rightarrow X$ in $\mathcal{C}$, while a morphism $(X, f) \rightarrow (Y, g)$ consists of an internal left adjoint $\varphi: X \rightarrow Y$ together with a 2-morphism $\alpha: \varphi \circ f \rightarrow g \circ \varphi$ i.e. a lax commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\varphi} & Y \\ f \downarrow & \alpha \nearrow & \downarrow g \\ X & \xrightarrow{\varphi} & Y \end{array}$$

Theorem A.6. [HSS17, Theorem 1.7.] *The $(\infty, 1)$ -category $\mathrm{End}^{\mathrm{tr}}(\mathcal{C})$ inherits a natural symmetric monoidal structure, such that the assignment taking $(X, f) \in \mathrm{End}^{\mathrm{tr}}(\mathcal{C})$ to the trace $\mathrm{tr}_{\mathcal{C}}(f) \in \Omega_{\mathcal{C}}$ refines to a symmetric monoidal functor*

$$\mathrm{Tr}_{\mathcal{C}}^\otimes: \mathrm{End}^{\mathrm{tr}}(\mathcal{C}) \longrightarrow \Omega_{\mathcal{C}}.$$

More explicitly, the functoriality of $\mathrm{Tr}_{\mathcal{C}}^\otimes$ works as follows. Given a map $(\varphi, \alpha): (X, f) \rightarrow (Y, g)$, denote by $\psi: Y \rightarrow X$ the internal right adjoint of φ . Then $\mathrm{Tr}_{\mathcal{C}}^\otimes$ takes the map (φ, α) to the natural transformation given by the composite

$$\mathrm{tr}_{\mathcal{C}}(f) \rightarrow \mathrm{tr}_{\mathcal{C}}(\psi \varphi f) \xrightarrow{\alpha} \mathrm{tr}_{\mathcal{C}}(\psi g \varphi) \simeq \mathrm{tr}_{\mathcal{C}}(\varphi \psi g) \rightarrow \mathrm{tr}_{\mathcal{C}}(g),$$

where the first and last map are given by the unit and counit of the adjunction $\varphi \dashv \psi$, respectively. Diagrammatically, these are also induced by the 2-cells in the following diagram

$$\begin{array}{ccccccc} \mathbb{1}_{\mathcal{C}} & \xrightarrow{\mathrm{coev}_X} & X \otimes X^\vee & \xrightarrow{f \otimes \mathrm{id}} & X \otimes X^\vee & \xrightarrow{\mathrm{ev}_X} & \mathbb{1}_{\mathcal{C}} \\ \parallel & \swarrow & \downarrow \varphi \otimes \psi^\vee & \alpha & \downarrow \varphi \otimes \psi^\vee & \swarrow & \parallel \\ \mathbb{1}_{\mathcal{C}} & \xrightarrow{\mathrm{coev}_Y} & Y \otimes Y^\vee & \xrightarrow{g \otimes \mathrm{id}} & Y \otimes Y^\vee & \xrightarrow{\mathrm{ev}_Y} & \mathbb{1}_{\mathcal{C}} \end{array} \quad (\text{A.1})$$

The $(-)^\vee: (\mathcal{C}^\diamond)^{\mathrm{op}} \rightarrow \mathcal{C}^\diamond$ induces a functor on $\mathrm{End}^{\mathrm{tr}}(\mathcal{C}^\diamond)$ in the obvious way. Note that in general $\mathrm{End}^{\mathrm{tr}}(\mathcal{C})^\diamond \neq \mathrm{End}^{\mathrm{tr}}(\mathcal{C}^\diamond)$.

Lemma A.7. *Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a unital lax symmetric monoidal functor of symmetric monoidal ∞ -categories. Let $X \in \mathcal{C}$ be an object such that $F(X) \in \mathcal{D}^\diamond$. Given any $\phi: X^\vee \rightarrow Y \in \mathcal{C}$ the induced map $F(\phi): F(X^\vee) \rightarrow F(Y)$ factors naturally through the comparison map*

$$\delta: F(X^\vee) \rightarrow F(X)^\vee$$

given by the adjoint of the composite

$$F(X^\vee) \otimes F(X) \rightarrow F(X^\vee \otimes X) \xrightarrow{F(\text{ev})} F(\mathbb{1}_{\mathcal{C}}) \simeq \mathbb{S}.$$

B Verdier duality

Throughout this section, fix a category $\mathfrak{B}$ which admits all finite limits and a symmetric monoidal functor

$$\mathcal{D}(-): \text{Span}(\mathfrak{B}) \longrightarrow \text{Pr}_{\text{st}}^{\text{L}},$$

where $\text{Span}(\mathcal{C})$ denotes the 1-category of spans in $\mathfrak{B}$ equipped with the cartesian monoidal structure. We write $\mathcal{D}(\text{pt}) = \mathcal{D}$ and note that $\mathcal{D}(\text{pt})$ refines to an object $\mathcal{D}(\text{pt}) \in \text{CAlg}(\text{Pr}_{\text{st}}^{\text{L}})$ and $\mathcal{D}(-)$ naturally refines to a symmetric monoidal functor

$$\mathcal{D}(-): \text{Span}(\mathfrak{B}) \longrightarrow \text{Mod}_{\mathcal{D}}(\text{Pr}_{\text{st}}^{\text{L}})$$

for which we abusively use the same notation. We also make the standing assumption that $\mathcal{D}$ is compactly generated and *rigid*, that is, an object of $\mathcal{D}$ is compact if and only if it is dualizable. For a map $f: X \rightarrow Y$ in $\mathfrak{B}$ we denote by

$$f^*: \mathcal{D}(Y) \longrightarrow \mathcal{D}(X)$$

the $\mathcal{D}(\text{pt})$ -linear left adjoint obtained by applying $\mathcal{D}(-)$ to the span

$$\begin{array}{ccc} & X & \\ f \swarrow & & \searrow \text{id} \\ Y & & X. \end{array}$$

Conversely, we denote by

$$f_!: \mathcal{D}(X) \longrightarrow \mathcal{D}(Y)$$

the $\mathcal{D}(\text{pt})$ -linear left adjoint obtained by applying $\mathcal{D}(-)$ to the span

$$\begin{array}{ccc} & X & \\ \text{id} \swarrow & & \searrow f \\ X & & Y, \end{array}$$

and denote the induced adjoints as

$$f^* \dashv f_* \quad f_! \dashv f^!.$$

Definition B.1. For any $X \in \mathfrak{B}$ and $\mathcal{E} \in \mathcal{D}(X)$, we write $\Gamma^c(X; \mathcal{E}) := X_! \mathcal{E} \in \mathcal{D}(\text{pt})$ and $\Gamma(X; \mathcal{E}) := X_* \mathcal{E}$. Moreover, for any map $f: X \rightarrow Y$ in $\mathfrak{B}$ we write $\omega_{X/Y} := f^! \mathbb{1}_{\mathcal{D}(Y)}$ and call this the *relative dualizing sheaf* of f . For $f = X \rightarrow \text{pt}$, being the terminal map, we also write $\omega_X := X^! \mathbb{1}$ and call this the (absolute) *dualizing sheaf* of X .

Remark B.2. The category $\mathcal{D}(X)$ inherits a natural presentably symmetric monoidal structure via the functors

$$\begin{aligned} X^*: \mathcal{D}(\text{pt}) &\longrightarrow \mathcal{D}(X) \\ \Delta^*: \mathcal{D}(X \times X) &\simeq \mathcal{D}(X) \otimes \mathcal{D}(X) \longrightarrow \mathcal{D}(X) \end{aligned}$$

Since $\mathcal{D}(X)$ is presentably monoidal, the functor

$$\mathcal{E} \otimes -: \mathcal{D}(X) \longrightarrow \mathcal{D}(X),$$

admits a right adjoint, which we denote by

$$\mathcal{H}\text{om}_{\mathcal{D}(X)}(\mathcal{E}, -): \mathcal{D}(X) \longrightarrow \mathcal{D}(X).$$

Such that, for any $\mathcal{F} \in \mathcal{D}(X)$ we have an equivalence

$$\Gamma(X; \mathcal{H}\text{om}(\mathcal{E}, \mathcal{F})) = X_* \mathcal{H}\text{om}_{\mathcal{D}(X)}(\mathcal{E}, \mathcal{F}) \simeq \text{map}_{\mathcal{D}(X)}(\mathcal{E}, \mathcal{F}) \in \mathcal{D}(\text{pt}).$$

Proposition B.3. *Let $f: X \rightarrow Y$ be a map in $\mathfrak{B}$. Then for any $\mathcal{E}, \mathcal{E}' \in \mathcal{D}(X)$ and $\mathcal{F} \in \mathcal{D}(Y)$, we have the following natural equivalences*

$$\begin{aligned} f_!(\mathcal{E} \otimes f^* \mathcal{F}) &\simeq f_!(\mathcal{E}) \otimes \mathcal{F} \\ \mathcal{H}\text{om}_{\mathcal{D}(Y)}(f_! \mathcal{E}, \mathcal{F}) &\simeq f_* \mathcal{H}\text{om}_{\mathcal{D}(X)}(\mathcal{E}, f^! \mathcal{F}) \\ f^! \mathcal{H}\text{om}_{\mathcal{D}(X)}(\mathcal{E}, \mathcal{E}') &\simeq \mathcal{H}\text{om}_{\mathcal{D}(Y)}(f^* \mathcal{E}, f^! \mathcal{E}'). \end{aligned}$$

Construction B.4. Since any $X \in \text{Span}(\mathfrak{B})$ is canonically self-dual, we also obtain a natural equivalence

$$\mathbb{V}: \mathcal{D}(X) \xrightarrow{\sim} \mathcal{D}(X)^\vee, \quad \mathcal{E} \mapsto X_!(- \otimes \mathcal{E})$$

associating to each $\mathcal{E} \in \mathcal{D}(X)$ the continuous functor given by the composite

$$\mathcal{D}(X) \xrightarrow{\pi_1^*} \mathcal{D}(X) \otimes \mathcal{D}(X) \xrightarrow{\text{id} \otimes \mathcal{E}} \mathcal{D}(X) \otimes \mathcal{D}(X) \xrightarrow{\Delta^*} \mathcal{D}(X) \xrightarrow{X_!} \mathcal{D}(X).$$

This comes together with, for any map $f: X \rightarrow Y$, identifications

$$\begin{aligned} f^* &\simeq \mathbb{V} \circ (f_!)^\vee \circ \mathbb{V} \\ f_! &\simeq \mathbb{V} \circ (f^*)^\vee \circ \mathbb{V}. \end{aligned}$$

In particular, composing with the internal dual in $\mathcal{D}(X)^\vee$, we obtain a functor

$$\mathbb{D}: \mathcal{D}(X) \xrightarrow{\mathbb{V}} \mathcal{D}(X)^\vee \xrightarrow{(\mathcal{H}\text{om}_{\mathcal{D}(X)}(-, \mathbb{1}))^\vee} (\mathcal{D}(X)^\vee)^{\text{op}} \xrightarrow{(\mathbb{V}^{-1})^{\text{op}}} \mathcal{D}(X)^{\text{op}}.$$

Unwinding the definitions, we see that $\mathbb{D}$ is given by

$$\mathbb{D} = \mathcal{H}\text{om}_{\mathcal{D}(X)}(-, \omega_X) = \mathcal{H}\text{om}_{\mathcal{D}(X)}(-, X^! \mathbb{1}).$$

Definition B.5. For any $X \in \mathfrak{B}$, we say that $\mathcal{E} \in \mathcal{D}(X)$ is *reflexive*, if the canonical map $\mathbb{D}^2(\mathcal{E}) \rightarrow \mathcal{E}$ is an equivalence.

Lemma B.6. *Let $f: X \rightarrow Y$ be a map in $\mathfrak{B}$ and $\mathcal{E} \in \mathcal{D}(X)$ and $\mathcal{F} \in \mathcal{D}(Y)$. Then, we have natural equivalences*

$$\begin{aligned} \mathbb{D}(f_! \mathcal{E}) &\simeq f_* \mathbb{D}(\mathcal{E}) \in \mathcal{D}(Y) \\ f^! \mathbb{D}(\mathcal{F}) &\simeq \mathbb{D}(f^* \mathcal{F}) \in \mathcal{D}(X). \end{aligned}$$

Proof. Indeed, the first equivalence is given by the composite

$$\mathbb{D}(f_! \mathcal{E}) = \mathcal{H}om(f_! \mathcal{E}, \omega_Y) \simeq f_* \mathcal{H}om(\mathcal{E}, f^! \omega_Y) \simeq f_* \mathcal{H}om(\mathcal{E}, \omega_X) \simeq f_* \mathbb{D}(\mathcal{E}),$$

while the second is obtained as

$$f^! \mathbb{D}(\mathcal{F}) = f^! \mathcal{H}om(\mathcal{F}, \omega_Y) \simeq \mathcal{H}om(f^* \mathcal{E}, \omega_X) = \mathbb{D}(f^* \mathcal{F}),$$

proving our claim. $\square$

Combining the two equivalences, we learn that for a sheaf $\mathcal{F} \in \mathcal{D}(Y)$ we have equivalences

$$\Gamma(X; f^! (\mathbb{D}(\mathcal{F}))) \simeq \Gamma(X; \mathbb{D}(f^* \mathcal{F})) = X_* \mathbb{D}(f^* \mathcal{F}) \simeq \mathbb{D}(X_! f^* \mathcal{F}) \simeq \Gamma^c(X; f^* \mathcal{F})^\vee.$$

In particular, for $Y = X$ and $f = \text{id}$ we have:

$$(X_! \mathcal{E})^\vee = \Gamma^c(X; \mathcal{E})^\vee \simeq \Gamma(X; \mathbb{D}(\mathcal{E})) = X_* \mathbb{D}(\mathcal{E}).$$

Put differently, this means that the functor $\mathcal{D}(X) \rightarrow \mathcal{D}(\text{pt})$ which takes $\mathcal{E}$ to $\Gamma^c(X; \mathcal{E})^\vee$ is represented by the dualizing sheaf ω_X . If X is additionally assumed to be proper, i.e. $X_! = X_*$, we may iterate this reasoning to see that

$$\Gamma(X; \mathbb{D}^2(\mathcal{E})) \simeq (\Gamma(X; \mathcal{E})^\vee)^\vee.$$

Corollary B.7. *For a proper map $f: X \rightarrow Y \in \mathfrak{B}$, the functor $f_!$ preserves reflexive sheaves.*

Proof. Since f is proper, we have $f_! = f_{\text{pt}}$ and so applying [Lemma B.6](#) twice we get for any reflexive $\mathcal{E} \in \mathcal{D}(X)$ equivalences

$$\mathbb{D}^2(f_! \mathcal{E}) \simeq \mathbb{D}(f_! \mathbb{D}(\mathcal{E})) \simeq f_! \mathbb{D}^2(\mathcal{E}) \simeq f_! \mathcal{E}.$$

Thus, $f_! \mathcal{E} \in \mathcal{D}(Y)$ is by definition reflexive. $\square$

Notice that, for any reflexive $\mathcal{F}, \mathcal{E} \in \mathcal{D}(X)$ we have

$$\mathcal{H}om(\mathcal{F}, \mathbb{D}(\mathcal{E})) \simeq \mathcal{H}om(\mathcal{F} \otimes \mathcal{E}, \omega_X) \simeq \mathbb{D}(\mathcal{F} \otimes \mathcal{E}).$$

Replacing $\mathcal{F}$ with $\mathbb{D}(\mathcal{F})$, we learn that we can compute the internal mapping sheaf between reflexive sheaves as the tensor product

$$\mathcal{H}om(\mathcal{F}, \mathcal{E}) \simeq \mathcal{H}om(\mathcal{F}, \mathbb{D}(\mathbb{D}(\mathcal{E}))) \simeq \mathbb{D}(\mathcal{F} \otimes \mathbb{D}(\mathcal{E})) \in \mathcal{D}(X).$$

Lemma B.8. *Let $f: X \rightarrow Y$ be a proper map in $\mathfrak{B}$ and let $\mathcal{E} \in \mathcal{D}(X)$. The following statements are equivalent:*

1. *The functor $f^*(-) \otimes \mathcal{E}$ has a left adjoint*
2. *The functor $f_!(\mathcal{E} \otimes -)$ has a continuous right adjoint*
3. *The functor $\mathcal{H}om(\mathcal{E}, f^!(-))$ has a right adjoint*

Moreover, in this case we have $\mathbb{D}^2(\mathcal{E}) \simeq \mathcal{E}$ and the adjunctions are given by

$$f_!(\mathbb{D}(\mathcal{E}) \otimes -) \dashv f^*(-) \otimes \mathcal{E} \simeq \mathcal{H}om(\mathbb{D}(\mathcal{E}), f^!(-))$$

and

$$\mathcal{H}om(\mathcal{E}, f^!(-)) \simeq f^*(-) \otimes \mathbb{D}(\mathcal{E}) \dashv f_!(\mathbb{D}(\mathcal{E})^\vee \otimes -)$$

Proof. Note that $f_!(\mathcal{E} \otimes -)$ always admits a right adjoint given by $\mathcal{H}om(\mathcal{E}, f^!(-))$, so conditions (2) and (3) are clearly equivalent. $\square$

Definition B.9. Let $f: X \rightarrow Y$ be a map in $\mathfrak{B}$. If $\mathcal{E} \in \mathcal{D}(X)$ satisfies any of the equivalent conditions of Lemma B.8, we call $\mathcal{E}$ *f -suave* or just *suave* if $Y = \text{pt}$ and f is the terminal map. We write $\mathcal{D}(X)^s$ for the full subcategory of $\mathcal{D}(X)$ spanned by the suave sheaves on X . Moreover, we call $X \in \mathfrak{B}$ suave if the unit $\mathbb{1}_X \in \mathcal{D}(X)$ is suave and *smooth* if additionally $\omega_X = X^!\mathbb{1}$ is dualizable.

In particular, suaveness ensures that the cohomology $\Gamma(X)$ is dualizable in $\mathcal{D}(\text{pt})$.

Proposition B.10. *Let $X \in \mathfrak{B}$ be proper and $\mathcal{E} \in \mathcal{D}(X)$ be suave. If $\mathbb{1}_{\mathcal{D}} \in \mathcal{D}^\omega$, the global sections $\Gamma(X; \mathcal{E})$ are a dualizable object of $\mathcal{D}(\text{pt})$.*

Proof. Since X is proper, the functor X^* is strongly continuous and hence preserves compact objects, hence the unit $X^*\mathbb{1} = \mathbb{1}_X \in \mathcal{D}(X)$ is compact. Moreover, since $\mathcal{E}$ is suave, the functor $X_!(- \otimes \mathcal{E})$ is strongly continuous and thus $\Gamma(X; \mathcal{E}) = X_!(\mathbb{1}_X \otimes \mathcal{E}) \in \mathcal{D}(X)$ is compact and hence dualizable, since $\mathcal{D}(\text{pt})$ is locally rigid. $\square$

For $X \in \mathfrak{B}$, denote by $\pi_i X \times X \rightarrow X$ the canonical projections and by $\Delta: X \rightarrow X \times X$ the diagonal. The *external tensor product* of two sheaves $\mathcal{E}, \mathcal{F} \in \mathcal{D}(X)$ is defined to be

$$\mathcal{E} \boxtimes \mathcal{F} := \pi_1^* \mathcal{E} \otimes \pi_2^* \mathcal{F} \in \mathcal{D}(X \times X).$$

Note that, we have a natural equivalence

$$\Delta^*(\mathcal{E} \boxtimes \mathcal{F}) \simeq \mathcal{E} \otimes \mathcal{F} \in \mathcal{D}(X).$$

Proposition B.11. *Let $\mathcal{E}, \mathcal{F} \in \mathcal{D}(X)$ be suave and denote by $\pi_i: X \times X \rightarrow X$ the two projections. Then, we have a natural equivalence*

$$\pi_1^* \mathbb{D}(\mathcal{E}) \otimes \pi_2^* \mathcal{F} \xrightarrow{\sim} \mathcal{H}om(\pi_1^* \mathcal{E}, \pi_2^! \mathcal{F}) \in \mathcal{D}(X \times X)$$

Corollary B.12. *If $\mathcal{E}, \mathcal{F} \in \mathcal{D}(X)$ are suave, we have a natural equivalence*

$$\Delta^!(\mathbb{D}(\mathcal{E}) \boxtimes \mathcal{F}) \simeq \mathcal{H}om(\mathcal{E}, \mathcal{F}) \in \mathcal{D}(X).$$

Proof. Follows by applying $\Delta^!$ to the equivalence of Proposition B.11 and observing that we have

$$\Delta^! \mathcal{H}om(\pi_1^* \mathcal{E}, \pi_2^! \mathcal{F}) \simeq \mathcal{H}om(\Delta^* \pi_1^* \mathcal{E}, \Delta^! \pi_2^! \mathcal{F}) \simeq \mathcal{H}om(\mathcal{E}, \mathcal{F})$$

by Proposition B.3 and the fact that $\pi_i \circ \Delta = \text{id}$. $\square$

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